ANALYSTLAB AFRICA

Machine Learning Track · Week 4

Consumer Complaint Volume Forecasting Prototype

Modelling report: data validation, forecasting, evaluation and limitations

Prepared by Umar Muhammad · CFPB complaints about Bank of America, May 2017 – Aug 2023

1. Executive summary

This project builds and tests a model that forecasts how many consumer complaints the CFPB receives about Bank of America each month. It uses 62,516 records, 75 complete months (May 2017 – Jul 2023), and a chronological, rolling-origin evaluation that never lets the model see the future.

- **Trend:** volume grew about 2.4 times between 2018 and 2022 (7,872 to 12,953 complaints a year). Seasonality is weak; the trend dominates.
- **Accuracy:** a typical 1 – 3 month forecast misses by about 8 – 13% of monthly volume (WAPE). Test months Jan – Jul 2023: 11.8% for the best model, 13.1% for "repeat last month". Without the July 2023 burst: 4.4% and 5.6%.
- **Main finding:** no method is reliably better than the simple baseline. The model that won validation (Ridge) lost to the baseline on the test months, and no improvement is statistically significant. The one robust result is that seasonal-naive forecasting is clearly worse.
- **Burst risk:** July 2023 saw 1,749 complaints (about 60% above the previous six months' typical level), concentrated on 11 – 15 July. No model anticipated it, and models that carried it forward over-forecast August by more than 120%.
- **Use:** treat forecasts as a planning range (about 1,100 – 1,500 complaints a month for Sep – Oct 2023), refreshed monthly, with burst monitoring. A forecast of complaint volume says nothing about misconduct or company performance.

2. Business context and responsible use

Stakeholders need to monitor complaint volumes and anticipate workload. A complaint is a consumer-reported issue; it does not establish misconduct or a regulatory violation. Recorded company responses are not proof that a complaint was resolved satisfactorily. This prototype forecasts the number of complaints received and makes no claim about the company's behaviour or future performance.

3. Data validation and preparation

The supplied workbook (62,516 rows, 12 columns) was preserved unchanged; all cleaning is done in memory and is reproducible from the notebook.

Check	Result	Decision
Complaint ID unique; duplicate rows	Unique; 0 duplicate rows (1,494 rows match another except the ID)	Kept: different IDs are different complaints
Blank Timely response?	1,494, exactly the In-progress complaints	Never counted as No; excluded from the timely-response denominator
Missing sub-issue / public response / sub-product	10,858 / 2,175 / 7	Labelled by meaning (not specified, no public response recorded)
Date validity	None forwarded before receipt; ranges	Valid

Check	Result	Decision
	match the brief	
Forwarding interval	Mean 1.22 d, median 0 d, 79.4% same day, 435 over 30 d, max 275 d	Kept; it is CFPB intake to forwarding, not company response time
Reporting months	76 consecutive months, none missing	No gap repair needed
Partial months	Aug 2023 covers 28 days (20 of 23 weekdays); May 2017 starts on day 1	Aug 2023 excluded from modelling, used as a separate check; May 2017 kept (no effect)
Zero-complaint product months	Student loan 46, payday/title loan 3	Genuine zeros, not missing data

Shared KPIs (consistent with the other tracks): total 62,516 complaints; timely-response 96.06% of valid records (58,619 Yes, 2,403 No), with 1,494 In-progress records excluded and reported separately; mean forwarding interval 1.22 days (median 0).

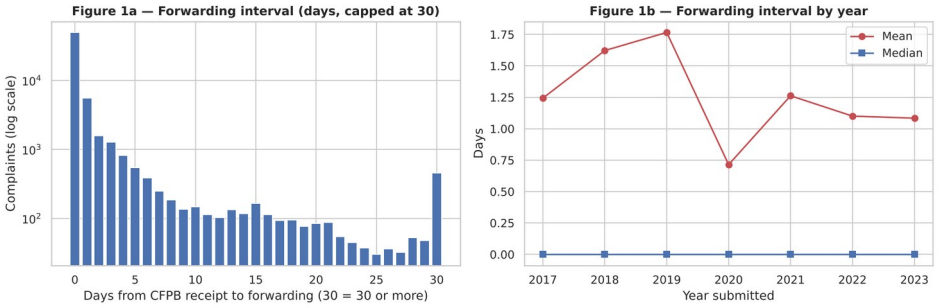


Figure 1 — CFPB forwarding interval: 79% same day, long tail

4. Historical trends

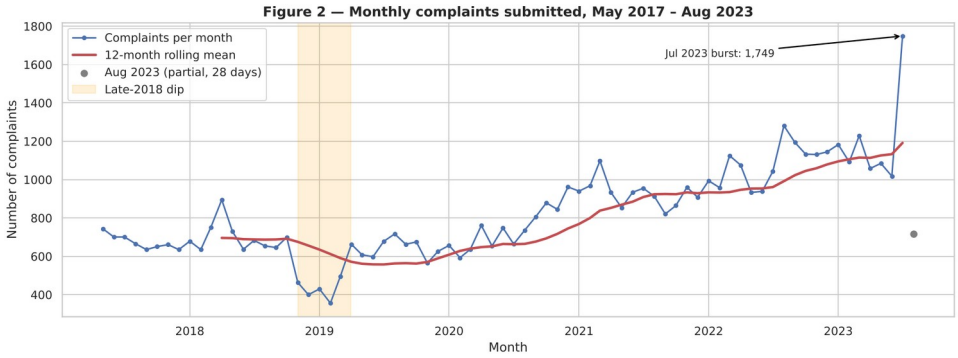


Figure 2 — Monthly complaints submitted, May 2017 – Aug 2023

- Trend strength 0.78 versus seasonal strength 0.16 (0 = none, 1 = dominant): volume is driven by a long-run trend, not a calendar pattern.
- A dip from November 2018 to March 2019 (as low as 356 a month) and a burst in July 2023 are visible. The data do not explain either, and no cause is assumed.
- The product mix shifts over time: mortgage complaints fall as a share, credit-card and credit-reporting complaints grow.

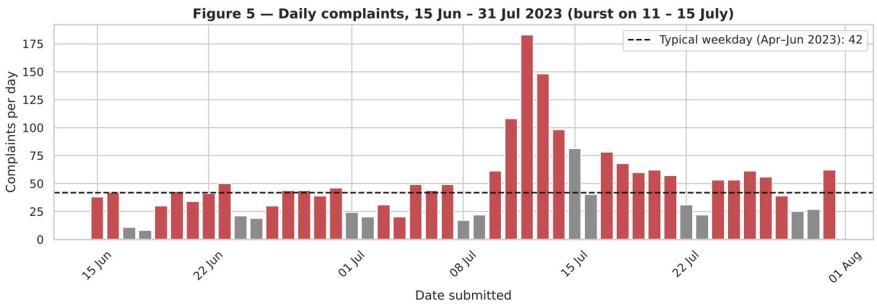


Figure 3 — The July 2023 burst: 11 – 15 July, peak 183 complaints in one day

5. Forecasting problem and validation design

Item	Definition
Target	Complaints submitted in a calendar month (by Date submitted), modelled as complaints per weekday and converted back using the target month's weekday count
Horizons	1 – 6 months ahead; planning focus 1 – 3 months
Features (history only)	Last month's rate, last 3 months, 6- and 12-month averages, recent change, month of year (sine/cosine), trend index
Validation	Rolling origin with an expanding window: every forecast is made from data up to its origin and nothing later. Validation = Jan – Dec 2022; Test = Jan – Jul 2023 (used once). Tuning only on validation.
Partial months	Aug 2023 excluded from training and evaluation; used as an extra out-of-sample check scaled to the weekdays actually covered
Metrics	MAE (average miss), RMSE (penalises large misses), WAPE (total absolute error ÷ total actual)

6. Models and results

Nine methods were compared: four baselines (naive, seasonal naive, 3-month average, 6-month median), exponential smoothing with damped trend, ARIMA (order chosen by AIC at every origin), ridge regression, gradient boosting, and an ensemble of the last three. Ridge penalty and boosting depth were chosen on validation months only.

6.1 Validation (2022), average over 1 – 3 month horizons

Method	MAE	RMSE	WAPE %
Ridge regression	82	112	7.6
Ensemble (ETS+Ridge+GBM)	91	110	8.4
6-month median	102	117	9.5
Gradient boosting	105	134	9.7
Naive	106	130	9.8
ARIMA	106	130	9.8
Exp. smoothing (damped)	107	126	9.9
3-month average	114	129	10.6
Seasonal naive	156	200	14.4

6.2 Test (Jan – Jul 2023), average over 1 – 3 month horizons

Method	MAE	RMSE	WAPE %	WAPE % excl. July
Gradient boosting	141	270	11.8	4.4
6-month median	145	253	12.1	5.8
3-month average	151	268	12.6	5.8
Ensemble (ETS+Ridge+GBM)	157	266	13.0	6.6
ARIMA	157	285	13.1	5.6
Exp. smoothing (damped)	158	280	13.1	6.0
Naive	157	285	13.1	5.6
Ridge regression	175	260	14.6	9.7
Seasonal naive	187	284	15.5	9.0

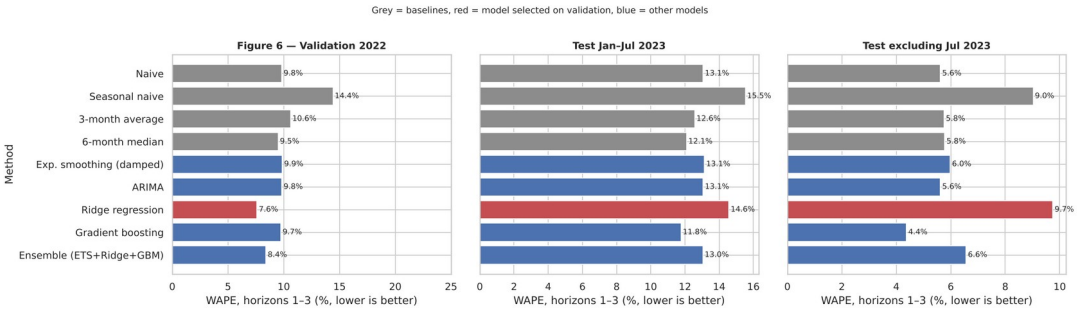


Figure 4 — Model comparison: validation, test, and test excluding July 2023

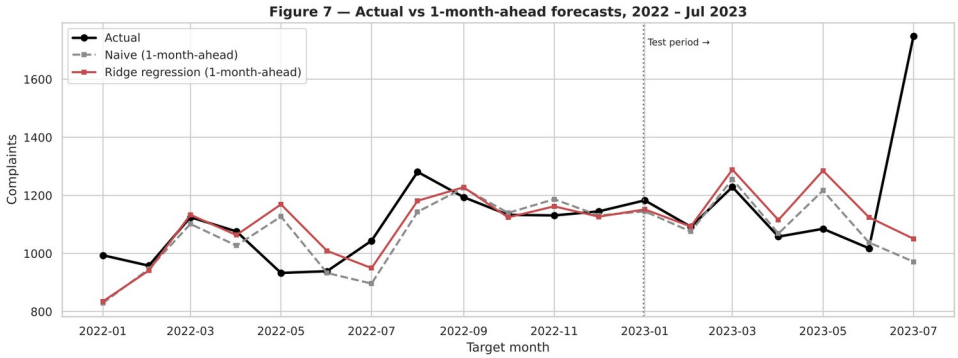


Figure 5 — Actual vs 1-month-ahead forecasts, 2022 – Jul 2023

Reading the results. Ridge was best on validation (7.6%) but worst of the non-seasonal methods on test (14.6%; 9.7% without July). ARIMA chose a plain random walk, ARIMA(0,1,0), at all 19 origins, so it equals the naive baseline. The ranking changes from one six-month window to the next:

Method	2022 Jan–Jun	2022 Jul–Dec	2023 Jan–Jul
Naive	10.5	9.2	13.1
6-month median	12.3	7.1	12.1
Ridge regression	10.6	5.0	14.6
Gradient boosting	7.3	11.8	11.8
Ensemble (ETS+Ridge+GBM)	9.4	7.5	13.0

Paired Wilcoxon tests against the naive baseline give p-values from 0.15 upwards for every method except seasonal naive, which is significantly worse ($p = 0.02$). With only 75 months, the extra complexity cannot be shown to beat "repeat last month". Because forecasts for neighbouring months are correlated, even these p-values are generous.

Error also grows with horizon (Figure 6), and sensitivity checks show that dropping May 2017 changes nothing, that using raw instead of weekday-adjusted counts raises the naive error from 11.1% to 11.8%, and that the July 2023 burst alone accounts for about 3 of the 11 percentage points of error.

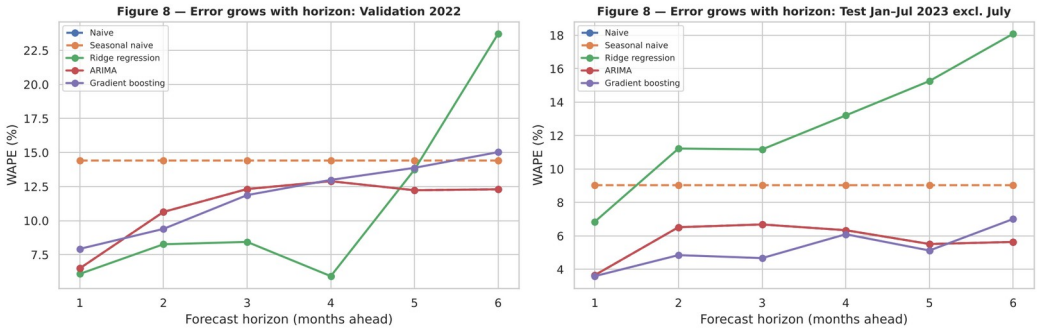


Figure 6 — Forecast error rises with the horizon

7. Error analysis

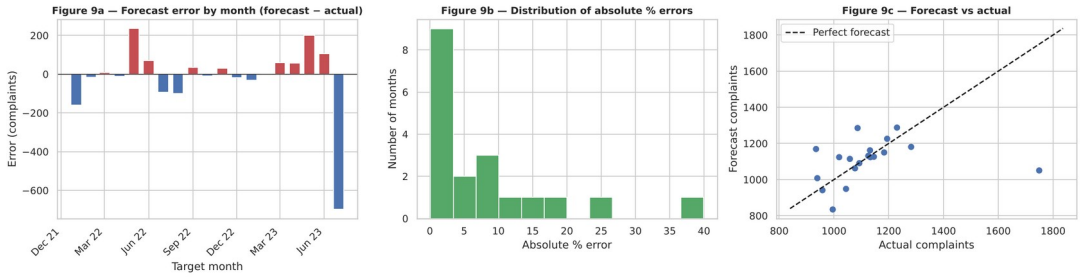


Figure 7 — Forecast errors by month, their distribution, and forecast vs actual

- The largest misses are turning points: July 2023 (actual 1,749, forecast 1,051), May 2022 (25% too high) and May 2023 (18% too high). Forecasts that lean on last month's level overshoot when volume falls back.
- Average error is -17.5 complaints (slightly low), driven by July 2023. Error autocorrelation is -0.06: no obvious unused pattern remains.

Partial-month check: August 2023

Method	Full-month forecast	Scaled to 20 weekdays	Actual (28 days)	Error %
Ridge regression	1843	1603	717	+123.5
Naive	1916	1666	717	+132.3
6-month median	1223	1063	717	+48.3
Gradient boosting	1917	1667	717	+132.5

August fell back to about 36 complaints per weekday (46 in June, 83 in July). Methods that lean on last month over-forecast by more than 120%; the 6-month median, which ignores a single burst, was much closer (+48%). A one-off burst must be flagged and not carried forward automatically.

8. Product-level forecasts

Six products average at least 30 complaints a month. One-month-ahead WAPE over the same validation and test months:

Product	Avg / month	Naive %	Exp. smoothing %	Ridge %	Gradient boosting %	Best
Checking or savings	327	13.2	13.3	14.1	13.9	Naive
Credit / prepaid card	213	12.2	11.3	12.0	11.5	Exp. smoothing (damped)
Credit reporting	101	10.3	10.4	11.7	12.6	Naive
Debt collection	36	20.4	20.0	21.9	22.8	Exp. smoothing (damped)
Money transfer / virtual currency	46	10.3	12.8	19.3	13.3	Naive
Mortgage	88	21.9	21.1	23.8	22.3	Exp. smoothing (damped)

Naive or exponential smoothing wins for every product; the machine-learning models do not help. Small products have the highest relative error because small counts are inherently noisy. Low-volume products (student loans: 46 zero months) are too sparse to forecast.

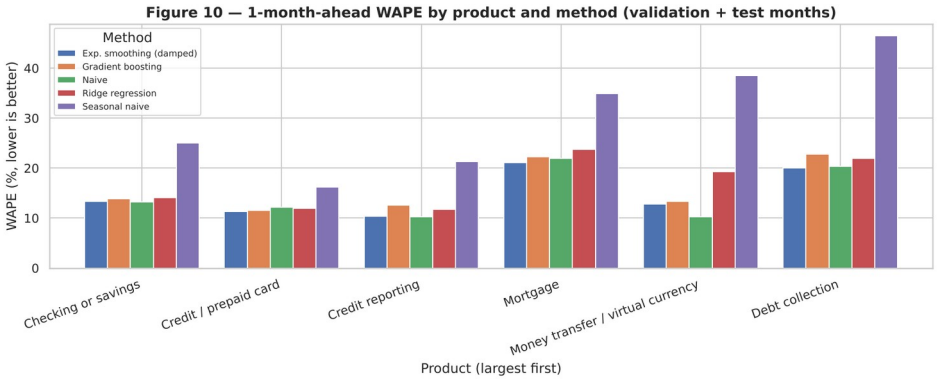


Figure 8 — One-month-ahead WAPE by product and method

9. Forecast for the coming months

Month	Forecast (selected model)	Rough 80% range	Robust-level scenario
2023-08 (full-month equivalent)	1843	1646 – 2060	1223
2023-09	1493	1198 – 1717	1117
2023-10	1478	1178 – 1595	1170

The as-is forecast is pulled upward by July's burst; the robust-level scenario treats July as a one-off. A sensible planning range for September – October is roughly 1,100 to 1,500 complaints. Ranges come from the model's own backtest errors (about 19 months per horizon), so they are rough. Horizons beyond 3 months were not shown to be reliable.

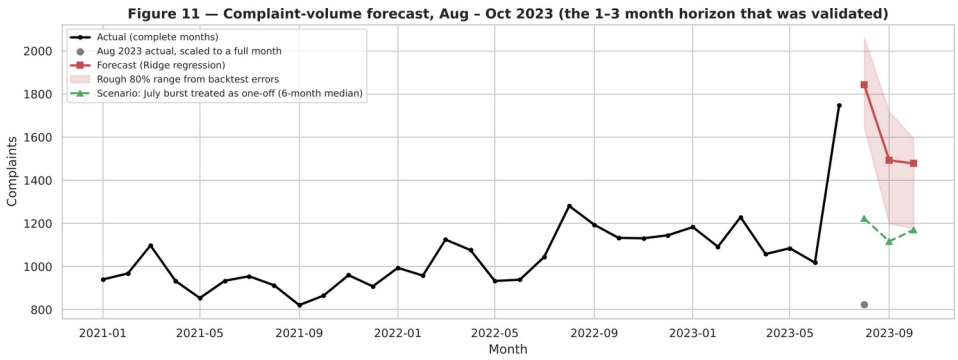


Figure 9 — Forecast for Aug – Oct 2023 with scenario and rough range

10. Limitations and recommendations

Limitations

- Short history (75 months) with a trend, a dip and a burst whose causes are not in the data.
- The validation winner did not generalise to the test months; differences between methods are small and not statistically significant.
- Only complaint counts are used; no external drivers are available or assumed.
- Interval estimates are rough, and forecasts beyond 3 months are unvalidated.
- A complaint is not misconduct, and the forecast does not describe company performance.

Recommendations

- Use forecasts as a planning range and refresh them monthly with the newest complete month.
- Keep the naive baseline in the loop and promote a more complex model only if it keeps beating it.
- Monitor daily counts for bursts (for example, more than twice the usual weekday level) so a one-off is not carried forward automatically.
- Forecast only the larger products (checking/savings, credit/prepaid card, credit reporting).
- Add external event indicators if stakeholders can supply them, and revisit the model as history grows.

Appendix — Reproducibility

File	Purpose
notebook/Complaint_Volume_Forecasting.ipynb	Full analysis with explanations, 11 figures
src/train_and_forecast.py	Re-trains the selected model from the raw workbook and writes forecasts (verified to match the notebook exactly)
models/model_recipe.json, complaint_forecaster.joblib	Model recipe, settings and series; the model is re-fitted at each origin
data/processed/*.csv	Monthly series, product series, backtest forecasts and metrics, future forecast
docs/DATA_QUALITY.md, docs/KPI_definitions.md	Data-quality log and shared KPI definitions